

Eduardo Alvarado

RESEARCH SCIENTIST IN HUMAN MOTION AND AVATAR SYNTHESIS - CV/ML/GRAPHICS

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Education

Ph.D. in Human Motion - Computer Vision, Machine Learning & Graphics

ÉCOLE POLYTECHNIQUE, LIX

Palaiseau, France

Oct. 2020 - Oct. 2023

- Thesis: Efficient Models for Human Locomotion and Interaction in Natural Environments.
- Supervisors: **Prof. Damien Rohmer** and **Prof. Marie-Paule Cani**.
- Funding: European Union's Horizon 2020 Research Programme. Marie Skłodowska-Curie Grant Agreement n. 860768 (CLIFE project).

M.Sc. in Embedded Systems and AI

ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

Freiburg, Germany

Apr. 2017 - Nov. 2019

- Robotics, Computer Vision, Machine Learning, Deep Learning and Reinforcement Learning.
- Thesis: Deep Multi-modal Learning for Autonomous Driving.
- Supervisors: **Prof. Joschka Bödecker** and **Prof. Abhinav Valada**.

B.Sc. in Electronics and Automation Engineering

UNIVERSIDAD CARLOS III - POZNAN UNIVERSITY OF TECHNOLOGY (ERASMUS)

Madrid, Spain

Sep. 2010 - Sep. 2014

- Digital/Analog Electronics, Robotics and Control Engineering.
- Thesis: Localization-based AR Framework for Smart-glasses.
- Supervisors: **Prof. Angel Garcia Crespo**.

Academic & Industry Experience

Postdoctoral Researcher - Computer Vision for Virtual Avatars and Human Motion

MAX PLANCK INSTITUTE FOR INFORMATICS

Saarbrücken, Germany

Jan. 2024 - Currently

- Research on 3D/4D motion analysis, synthesis and 3D avatar generation. In-depth understanding of GenAI and multi-modal foundation models.
- Supervision of Ph.D. and M.Sc. students, leading research projects across institutions and defining technical directions.
- Advised by: **Prof. Christian Theobalt** and **Dr. Marc Habermann**.
- **Parental Leave Breaks:** 7 months (Sep. 2024 — Apr. 2025) and 5 months (Aug. 2026 — Dec. 2026).

Ph.D. Intern - Authoring Tools for Character Animation

UBISOFT LA FORGE

Bordeaux, France

Mar. 2023 - Jun. 2023

- Research on ML techniques for production-ready FK/IK algorithms for game development tools.

Research Scientist - AI for Material Rendering

INSTALOD GMBH

Stuttgart, Germany

Jun. 2020 - Sep. 2020

- Research on zero-shot AI methods for 3D material generation from single RGB images (smartphones).

M.Sc. Thesis - Deep Multi-modal Learning for Autonomous Driving

ROBERT BOSCH GMBH

Renningen, Germany

May 2019 - Nov. 2019

- Research on sensor fusion for Autonomous Driving. Development of AI frameworks for Radar and RGB data acquisition and manipulation.

Research Assistant - Autonomous Intelligent Systems

ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

Freiburg, Germany

Sep. 2017 - Feb. 2019

- TurtleBot3 Waffle hardware/software setups for teleoperation, SLAM and indoor mapping.

M.Sc. Intern - Dev-Tools Engineer for Autonomous Driving

ZF FRIEDRICHSHAFEN AG

Friedrichshafen, Germany

Apr. 2018 - Oct. 2018

- Development of computer vision ROS tools for vehicle data.

Business Strategy Analyst and QA Engineer for Robotics

BQ MUNDO READER S.L.

Madrid, Spain

Nov. 2014 - May 2016

- Responsible for Strategy Planning and Quality Monitoring. In-depth knowledge of Consumer Robotics. Lecturer for Robotics Workshops.

Languages

Spanish Native **English** Fluent **German** Fluent **French** Intermediate

Skills

Programming	Python, C#, C++, 图形学	Game/3D	Unity, Unreal Engine, Blender	Robotics	ROS, Gazebo
AI	PyTorch, Tensorflow	OS	Windows, Linux	SW-Dev	CMake, Visual Studio, PyCharm, Git

Academic Activities

Research Visitor

UNIVERSITAT POLITÈCNICA DE CATALUNYA

Barcelona, Spain

Sep. 2022 - Dec. 2022

- Topic: Crowds Authoring in Natural Environments. Supervisor: Prof. Nuria Pelechano.

Research Visitor

CYENS CENTRE OF EXCELLENCE

Nicosia, Cyprus

Jul. 2021 - Oct. 2021

- Topic: Character Animation with focus on Motion Capture and Motion Matching. Supervisor: Prof. Yiorgos Chrysanthou.

Teaching Assistant for INF633 - Advanced 3D Graphics

ÉCOLE POLYTECHNIQUE

Palaiseau, France

Oct. 2020 - Oct. 2023

- Topic: Ecosystems Authoring, Procedural Animations, AI for Behavior Planning.

Reviewer / Committee

SIGGRAPH, SIGGRAPH ASIA, IEEE VR, TVCG, EUROGRAPHICS (PC), CEIG (PC), CASA (PC)

Worldwide

Oct. 2020 - Currently

Selected Publications

PEER-REVIEWED JOURNAL AND CONFERENCE ARTICLES

SOMA: From Surface Observations to Muscle Anatomy (🔄📄📺)

EDUARDO ALVARADO, EMILY KIM, GERRIT NOLTE, FRIEDEMANN RUNTE, MARIO BOTSCH, MARC HABERMANN, CHRISTIAN THEOBALT

European Conference on Computer Vision (ECCV), 2026

SoleSense: Insole-guided Stability for Diffusion-driven Humanoids

SIDHARTH SURESH NAIR, **EDUARDO ALVARADO**, CHRISTIAN THEOBALT

(Under Review), 2026

Towards Real-World Wearable Motion Reconstruction (🔄📄📺)

ANDREA BOSCOLO CAMILETTO, RISHABH DABRAL, **EDUARDO ALVARADO**, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT

European Conference on Computer Vision (ECCV), 2026

Step2Motion: Locomotion Reconstruction from Pressure Sensing Insoles (🔄📄📺)

JOSE LUIS PONTON, **EDUARDO ALVARADO**, LIN GENG FOO, NURIA PELECHANO, CARLOS ANDUJAR, MARC HABERMANN

Eurographics, 2026

FRAME: Floor-aligned Representation for Avatar Motion from Egocentric Video (🔄📄📺)

ANDREA BOSCOLO CAMILETTO, JIAN WANG, **EDUARDO ALVARADO**, RISHABH DABRAL, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT

Computer Vision and Pattern Recognition (CVPR), 2025

BimArt: A Unified Approach for the Synthesis of 3D Bimanual Interaction with Articulated Objects (🔄📄📺)

WANYUE ZHANG, RISHABH DABRAL, VLADISLAV GOLYANIK, VASILEIOS CHOUTAS, **EDUARDO ALVARADO**, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT

Computer Vision and Pattern Recognition (CVPR), 2025

TRAIL: Simulating the Impact of Human Locomotion on Natural Landscapes (🔄📄📺)

EDUARDO ALVARADO, OSCAR ARGUDO, DAMIEN ROHMER, MARIE-PAULE CANI, NURIA PELECHANO

Computer Graphics International (CGI), 2024

Generating Upper-Body Motion for Real-Time Characters Making their Way through Dynamic Environments (🔄📄📺)

EDUARDO ALVARADO, DAMIEN ROHMER, MARIE-PAULE CANI

Proceedings of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), 2022 —  **Best Paper Honorable Mention Award**

Real-Time Locomotion on Soft Grounds With Dynamic Footprints (🔄📄📺)

EDUARDO ALVARADO, CHLOÉ PALIARD, DAMIEN ROHMER, MARIE-PAULE CANI

Frontiers in Virtual Reality, Frontiers, 2022, 3, (10.3389/frvr.2022.801856)

A Survey on Reinforcement Learning Methods in Character Animation (📄)

ARIEL KWIATKOWSKI, **EDUARDO ALVARADO**, VICKY KALOGEITON, KAREN C. LIU, JULIEN PETTRÉ, MICHIEL VAN DE PANNE, MARIE-PAULE CANI

Eurographics 2022, Computer Graphics Forum, Wiley, 2022, pp.1-27. (10.1111/cgf.14504)

Interests

I'm passionate about skiing and try to explore a new destination each year with friends and family. I stay actively engaged with the latest developments in robotics and AI, often sharing my curiosity with my son to spark his interest. In my spare time, I work on building my own robotics startup, alongside learning piano, doing DIY projects, and exploring game-dev.